

APPENDIX E

PERFORMANCE-BASED DESIGN PROCEDURES FOR FIRE EFFECTS ON STRUCTURES

E.1 SCOPE

This appendix is not a mandatory part of the standard. It provides procedures for performance-based design and evaluation of structures for fire conditions that result in fire-induced effects on a structure's members and connections. The use of performance-based structural fire design procedures constitutes an alternative methodology to meet project design requirements, as is permitted by Section 1.3.7 and the alternative materials, design, and methods of construction provision in the building codes. This appendix does not provide for standard fire resistance design with prescriptive methods, nor does it address explosions.

E.2 DEFINITIONS

The following definitions apply to the information presented in this appendix.

FIRE: An oxidation process that results in the burning of flammable materials and produces heat.

FIRE EFFECTS: Thermal and structural response caused by fire exposure and subsequent cooling.

FIRE EXPOSURE: The extent to which materials, products, or assemblies are subjected to the conditions created by fire.

FIRE RESISTANCE: The ability of a material, product, or assembly to withstand fire or provide protection from it for a period of time.

FUEL LOAD: The total quantity of combustible contents within a building, space, or area expressed either as total energy or equivalent mass.

HEAT TRANSFER: The exchange of thermal energy caused by a temperature difference.

PERFORMANCE-BASED STRUCTURAL FIRE DESIGN: The explicit design of structural members and connections to satisfy performance objectives for structural design fires.

STANDARD FIRE RESISTANCE DESIGN: The selection of fire-resistive assemblies to meet code requirements for structural fire resistance (also known as prescriptive design). The rating of a fire-resistive assembly is based on its performance in standard fire testing.

STRUCTURAL DESIGN FIRE: A fire that has the potential to affect the integrity and stability of a structure that is used for the design and evaluation of the structure.

THERMAL BOUNDARY CONDITION: The temperature and/or heat flux to which an assembly or the structure is subjected during or after fire exposure based upon the radiative and convective heating and/or cooling conditions at exposed surfaces.

THERMAL INSULATION: A material or medium that reduces the heat transfer between objects in thermal contact or in range of radiative or convective influence.

THERMAL RESPONSE: The temperature distribution of members and connections when exposed to thermal boundary conditions.

THERMAL RESTRAINT: A condition in which thermal expansion or contraction of structural members is resisted by forces external to the members. The level of restraint depends on the adjacent framing and connection details.

E.3 GENERAL REQUIREMENTS

Performance-based structural fire design and evaluation procedures shall comply with the requirements of Section 1.3.1.3.

Design and evaluation of structures for fire effects shall include the following tasks: identify performance objectives, quantify the fuel load, identify and evaluate structural design fires, determine temperature histories of structural members and connections, and determine the response of the structure. These tasks shall include evaluating the heating and subsequent cooling of the structure under fire exposure, as appropriate.

The structural response shall be evaluated against limit states based on the performance objectives. The analyses shall account for temperature-dependent material properties, boundary conditions, and thermally induced failure modes and shall evaluate structural stability, strength, deformation, and load path continuity.

E.4 PERFORMANCE OBJECTIVES

Performance objectives shall be expressed as quantifiable metrics for the design and evaluation of structural response to structural design fires. Performance objectives for structural integrity, including strength, stiffness, and stability, shall meet the minimum criteria specified in Section E.4.1. Additional project-specific performance objectives shall meet the requirements in Section E.4.2.

E.4.1 Structural Integrity. Structural integrity shall be provided for buildings and other structures subject to structural design fires such that the structural system remains stable with a continuous load path to the extent necessary to ensure occupant life safety.

The performance of the structural system under structural design fires shall allow building occupants to travel safely to refuge areas within the building or exit the building to a public way. Structural support of building egress routes shall be maintained for a period of time necessary to ensure a safe and complete evacuation of building occupants. Structural support of building refuge areas shall be maintained throughout the heating and cooling of the structure.

E.4.2 Project-Specific Performance Objectives. Buildings and other structures shall be designed to meet project-specific

performance objectives required by the owner, Authority Having Jurisdiction, or applicable building code in addition to those in Section E.4.1.

E.5 THERMAL ANALYSIS OF FIRE EFFECTS

The thermal response of structural members and connections during and after structural design fires shall be determined for input to the structural analyses of fire effects.

E.5.1 Fuel Load. The fuel load shall be quantified for use in the evaluation of structural design fires.

E.5.2 Structural Design Fires. Structural design fires shall be identified and used to develop time-dependent thermal boundary conditions for use in heat transfer analyses.

E.5.3 Heat Transfer Analysis. The temperature history of structural members and connections shall be determined using heat transfer analysis methods based on the time-dependent thermal boundary conditions for structural design fires.

Temperature-dependent thermal properties of materials comprising the structural system and thermal insulation shall be used in heat transfer analyses to determine the thermal response. It shall be permissible to use constant thermal property values if they yield conservative results.

E.6 STRUCTURAL ANALYSIS OF FIRE EFFECTS

Structural analyses shall include portions of the structural system that are subject to fire effects from the structural design fires as determined in Section E.5, with consideration of unheated portions of the structural system that provide thermal restraint. It is permissible for the analyses to consider alternate load paths that are capable of being maintained following structural damage or degradation caused by fire effects.

E.6.1 Temperature History for Structural Members and Connections. Temperature histories for structural members and connections shall be determined from thermal analysis of structural design fires and shall be used to analyze fire effects on structural performance.

E.6.2 Temperature-Dependent Properties. Temperature-dependent properties of structural materials shall be used to determine the performance of structural members and connections subject to structural design fires.

E.6.3 Load Combinations. Load combinations in Section 2.5 for extraordinary events shall be used for analysis of fire effects and shall include time- and sequence-dependent effects. To check the residual capacity of a structure damaged by a structural design fire, the approach identified in Section 2.5.2.2 shall be used.

COMMENTARY TO STANDARD ASCE/SEI 7-16

This commentary is not a part of the ASCE Standard *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*. It is included for information purposes.

This commentary consists of explanatory and supplementary material designed to assist local building code committees and regulatory authorities in applying the recommended requirements. In some cases it will be necessary to adjust specific values in the standard to local conditions. In others, a considerable amount of detailed information is needed to put the

provisions into effect. This commentary provides a place for supplying material that can be used in these situations and is intended to create a better understanding of the recommended requirements through brief explanations of the reasoning employed in arriving at them.

The sections of the commentary are numbered to correspond to the sections of the standard to which they refer. Because it is not necessary to have supplementary material for every section in the standard, there are gaps in the numbering in the commentary.

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